BL Weather Report

Technical Guide

Rainfall Monitoring & NDVI Analysis — Methodology & Calculation Reference

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1. Overview

The **BL Weather Report** workflow is a data-to-report pipeline built for Big Life Foundation's rainfall monitoring programme in the Amboseli ecosystem. It ingests rainfall gauge observations from **EarthRanger**, derives vegetation condition from **Google Earth Engine** NDVI time-series, and delivers a set of charts, an interactive rainfall map, per-ranch NDVI trend charts, and a print-ready Word report.

The workflow is implemented as an **Ecoscope Workflow** (YAML spec) and executes within the ecoscope-workflows runtime. NDVI analysis is fanned out per ranch using the mapvalues directive, grouping by the R_NAME column of the Amboseli conservancy boundaries GeoDataFrame.

The workflow delivers the following outputs per run:

Category	Outputs
Rainfall charts	Overall rainfall bar chart (per station), monthly stacked bar chart (per station × month)
Rainfall map	Interactive proportional-symbol map of readings per station
NDVI charts	Per-ranch historic NDVI time-series with min/max/mean band (one chart per conservancy)
Data files	Aggregated station readings (GeoPackage), monthly readings (GeoPackage), per-ranch NDVI CSV
Report	Word document: Big Life weather report template populated with all charts and map images
Dashboard	Metadata dashboard (run details, time range, groupers)

2. Dependencies & Prerequisites

2.1 EarthRanger Connection

All rainfall data is sourced from an **EarthRanger** instance via set_er_connection. The workflow fetches events of type rainfall_rep using get_events. raise_on_empty: true is set, so the workflow will halt with an error if no rainfall events are found for the period — distinguishing a data-availability problem from a genuine zero-reading period.

The EarthRanger connection is also used to look up the current user's full name (get_current_user → get_user_full_name), which is inserted into the *Generated by* field of the Word report.

2.2 Google Earth Engine Connection

A **Google Earth Engine (GEE)** service account is required for NDVI analysis. set_gee_connection stores the project handle, which is passed to calculate_ndvi_range for each ranch ROI. GEE credentials must be stored as a JSON private key; see the GEE Setup Guide for instructions.

2.3 Static Geospatial Layers

Three GeoPackage files are downloaded automatically from Dropbox at run time. All three are reprojected to **EPSG:4326 (WGS 84)** before use.

File	Layer used for	Key columns
amboseli_ranch_conservancies_layers.gpkg	Land-use fill polygons on map	land_use (5 categories)
amboseli_group_ranch_boundaries_x_electric_fence.gpkg	Ranch boundaries + electric fence lines on map	land_use (Ranch boundaries, Electric Fence)
amboseli_group_ranch_boundaries.gpkg	ROI polygons for per-ranch NDVI calculation	R_NAME (ranch / conservancy name)

2.4 Base Map Tile Layers

Every map uses two stacked tile services:

Layer	URL (abbreviated)	Opacity	Max zoom
ESRI World Hillshade	arcgisonline.com/…/World_Hillshade/…	1.0	20
ESRI World Street Map	arcgisonline.com/…/World_Street_Map/…	0.15	20

2.5 Grouping Strategy

The grouper is fixed to R_NAME — the ranch / conservancy name field in the Amboseli boundaries GeoPackage. This column drives the split_groups fan-out that produces one NDVI chart per ranch. The temporal index added to rainfall events also uses this grouper.

3. Rainfall Data Pipeline

3.1 Event Fetch — rainfall_rep

get_events queries EarthRanger for events of type rainfall_rep within the analysis time range. include_details: true ensures the event_details JSON blob is returned, which carries the actual gauge readings. Only events with a point geometry are included (include_null_geometry: false).

Columns retained per event:

- id, time, event_type, event_category, reported_by, serial_number, geometry, created_at, event_details

3.2 JSON Normalisation

normalize_json_column flattens the nested event_details JSON object into individual columns. The resulting columns follow the pattern event_details__<key>. skip_if_not_exists: true makes this step a no-op if the column is already flat, ensuring robustness across EarthRanger versions. sort_columns: true reorders the output columns alphabetically for deterministic downstream access.

3.3 Column Renaming

transform_columns renames the three key fields from their raw JSON-flattened names to shorter working names:

Raw column name	Working name	Description
event_details__rainfallrep_raingauge_digital	digital_readings	Reading from the digital rain gauge
event_details__rainfallrep_raingauge_manual	manual_readings	Reading from the manual rain gauge
event_details__raingauge_stations	raingauge_stations	Name of the gauge station

skip_missing_rename: true suppresses errors if any of these columns are absent in a given run (e.g. if the EarthRanger event form schema differs). required_columns is still declared, so the workflow will warn on missing fields without halting.

3.4 Combined Readings — Digital / Manual Fallback

create_combined_column merges the digital and manual gauge readings into a single overall_readings column using a priority fallback strategy:

- **Primary:** digital_readings — used when available and non-null
- **Fallback:** manual_readings — used when the digital reading is absent
- **fill_remaining: 0** — rows where both are null receive a zero reading

This design reflects field practice: digital gauges are preferred for accuracy, but rangers fall back to manual readings when the digital gauge malfunctions or has not yet been read.

3.5 Data Cleaning

Three sequential cleaning steps ensure numeric integrity:

Task	What it does
replace_empty_strings	Replaces blank raingauge_stations values with 'Unknown', preventing empty-string grouping keys downstream
convert_to_float	Casts overall_readings to float; non-parseable values are coerced to NaN then filled with 0
fill_numeric_nulls	Fills any remaining NaN in overall_readings with 0 — ensures the aggregation step receives no nulls

3.6 Aggregation by Station

groupby_aggregate collapses all events to one row per gauge station:

Column	Aggregation	Result
overall_readings	sum	Total rainfall in mm recorded at each station for the period
geometry	first	Station point geometry (all readings for a station share the same location)

A parallel monthly aggregation branch (group_month_rain) groups by month_name × raingauge_stations and is used exclusively for the monthly stacked bar chart.

4. Chart Outputs

4.1 Overall Rainfall Bar Chart

draw_bar_chart renders a simple bar chart of total rainfall (mm) per station for the full analysis period.

Property	Value
X axis	raingauge_stations — one bar per station
Y axis	overall_readings (sum) — total mm
Bar colour	#6495ed (cornflower blue)
Plot background	#f5f5f5 (light grey)
Legend	Hidden (showlegend: false)
Bar gap	0.1

4.2 Monthly Stacked Bar Chart

draw_custom_stacked_bar_chart renders a stacked bar chart showing month-by-month rainfall contributions at each station. Each stack segment represents one calendar month.

Month name derivation

return_month_name extracts the month name string (e.g. 'January') from the time column and appends it as month_name. ordered_month_list returns the canonical January–December ordering, which is passed to stack_order to ensure consistent month sequencing regardless of which months have data.

Month colour assignment

apply_color_map maps each month_name value to a shade from the **Blues_r** sequential colormap — earliest months are darker blue, later months lighter — providing an intuitive temporal gradient within each stacked bar.

Property	Value
X axis	raingauge_stations — one grouped bar per station
Y axis	overall_readings (sum per month per station)
Stack column	month_name
Stack order	January → December (canonical)
Colour column	colors (Blues_r colormap)
Plot background	#f5f5f5
Legend	Visible

5. Rainfall Map

An interactive proportional-symbol map where each gauge station is represented by a circle whose radius scales with total rainfall, overlaid on land-use polygons and ranch boundaries.

5.1 Geometry Filtering

Before map layer creation, two cleaning steps remove problematic geometries from the aggregated station GeoDataFrame:

- **exclude_geom_outliers** — removes stations whose location is a statistical outlier (Z-score > 3 on coordinate values). This eliminates stray GPS artefacts that would skew the map extent.
- **drop_null_geometry** — removes any remaining rows with null geometry before DeckGL layer creation, preventing rendering errors.

5.2 Proportional Symbol Layer

create_scatterplot_layer draws one circle per station. The radius is driven by the overall_readings column, making larger rainfall totals visually larger:

Property	Value	Effect
get_fill_color	[100, 149, 237] #6495ed	Cornflower blue fill — matches bar chart colour
get_line_color	[70, 119, 207]	Slightly darker outline
get_radius	overall_readings	Radius driven by the rainfall total
radius_scale	0.044	Scales raw mm values to screen pixels
radius_min_pixels	5	Minimum circle size — stations with zero rainfall remain visible
radius_max_pixels	20	Caps circle size to prevent overlap at high values
radius_units	pixels	Screen-space radius, zoom-independent
opacity	85 %	Slightly transparent to show polygon layers beneath
stroked	true	Outline drawn

5.3 Text Label Layer

A text layer renders each station's total rainfall value (in mm, rounded to the nearest integer) directly on the map:

- overall_readings is converted to int then str before passing to the text layer
- Font: **Calibri Bold**, 65 px base size, 90–180 px screen clamp
- get_pixel_offset: [-1, -5] — nudges the label slightly above the circle centre
- Background padding: [-3, -20] — tight horizontal / loose vertical clearance
- Text colour: black (0, 0, 0, 255); billboard mode (always faces the viewer)
- pickable: false — labels do not trigger hover tooltips

5.4 Land-use Fill Layers

The Amboseli ranch conservancies layer is split by land_use column and styled independently:

Land-use Category	Fill / Line Colour	Opacity
Rangeland	RGB(163, 156, 145) — #a39c91	75 %
Rangeland and settlement	RGB(255, 160, 122) — #ffa07a	75 %

Land-use Category	Fill / Line Colour	Opacity
Settlement and agriculture	RGB(47, 79, 79) — #2f4f4f	75 %
Conservancies	RGB(143, 188, 139) — #8fbc8b	75 %
National parks and reserves	RGB(77, 102, 0) — #4d6600	75 %

5.5 Ranch Boundaries and Electric Fence

The boundaries layer is split by land_use and styled as line layers (no fill):

Feature	Colour	Width / Opacity	Notes
Ranch boundaries	RGB(0, 0, 0) — black	0.95 / 55 %	Thin black outlines delineating each group ranch
Electric Fence	RGB(45, 98, 177) — #2d62b1	5.5 px / 100 %	Bold blue lines marking the electric fence alignment

The map view state is set by view_state_deck_gdf using the rainfall station GeoDataFrame extent (pitch 0, bearing 0), constraining the initial view to the area of recorded measurements.

6. NDVI Analysis

Vegetation condition is assessed per ranch using the Normalised Difference Vegetation Index (NDVI) derived from MODIS satellite imagery. The analysis produces one historic time-series chart per conservancy, showing how current-period NDVI compares against the multi-year historic range.

6.1 ROI Split by Ranch

split_groups partitions the Amboseli conservancy boundaries GeoDataFrame by the R_NAME column, producing one GeoDataFrame per ranch. Each GeoDataFrame is used as the roi (region of interest) for a separate calculate_ndvi_range call via mapvalues.

dataframe_column_first_unique_str extracts the ranch name from each split GeoDataFrame. Ranch names are then zipped with the corresponding NDVI chart HTML via zip_groupbykey, so each persisted file is suffixed with the correct ranch name.

6.2 NDVI Method

The user selects one of two MODIS-based NDVI methods at run time via the ndvi_method string variable:

Method	Source	Resolution	Temporal resolution	Best for
MODIS MYD13A1 16-Day Composite	Pre-calculated NDVI	500 m	16-day composites	Phenology studies; quality-filtered best-pixel values; ~0.025 accuracy; may saturate in dense canopies

Method	Source	Resolution	Temporal resolution	Best for
MODIS MCD43A4 Daily NBAR	NIR/Red band ratio	500 m	Daily (view-angle corrected)	Higher temporal resolution; consistent measurements across acquisitions; more susceptible to cloud gaps

Default is **MODIS MYD13A1 16-Day Composite**. baseline_time_range is set to null and image_size to 1 000 000 000 (effectively unlimited), both pending support for a baseline time range parameter in the web form.

> **Note:** NDVI data is persisted per ranch as a CSV (persist_df_wrapper with sanitize: true, which replaces special characters in column names). The skipif: never condition on this task ensures the CSV is always written even when upstream tasks trigger skip conditions.

6.3 Historic Time-series Chart

draw_historic_timeseries renders the per-ranch NDVI chart with three visual layers:

Visual element	Column	Style
Current NDVI line	NDVI	Default plotly line style; labelled 'NDVI'
Historic min-max band	min / max	Filled area between min and max; fill: rgba(144, 238, 144, 0.3) — light green; line: rgba(144, 238, 144, 0.8)
Historic mean line	mean	Default plotly line style; labelled 'Historic Mean'

The time axis uses the img_date column. The min-max band is drawn as a filled area between two mode: lines traces, giving a visual envelope of multi-year NDVI variability. When the current NDVI line falls outside this envelope it indicates an anomalously wet or dry condition relative to the historical baseline.

7. Word Report

The final deliverable is a **Word document** (big_life_weather_report.docx) assembled from a Dropbox-hosted template populated with all generated charts and the rainfall map.

7.1 Template Download

The template file (big_life_weather_report_template.docx) is downloaded automatically from Dropbox at run time (fetch_and_persist_file, up to 3 retries). overwrite_existing: false skips the download on subsequent runs if the file is already cached, saving bandwidth.

7.2 Author Attribution

The workflow looks up the currently authenticated EarthRanger user via get_current_user and resolves their display name with get_user_full_name. This name is passed as generated_by to generate_weather_report, so the report is automatically attributed to the ranger who ran it without requiring manual input.

7.3 Report Generation

generate_weather_report populates the template context and inserts all map and chart images into the document:

Parameter	Value	Description
template_path	downloaded .docx	Big Life weather report template
output_dir	ECOSCOPE_WORKFLOWS_RESULTS	Output directory
generated_by	EarthRanger user full name	Author field in the report
box_height	3.12 inches	Image placeholder height
box_width	5.34 inches	Image placeholder width
time_period	analysis time_range	Report period displayed in the header
filename	big_life_weather_report.docx	Output filename
validate_images	true	Checks that all expected image files exist before populating placeholders

Images are sourced from the PNG screenshots produced by html_to_png tasks at 2× device scale factor. The rainfall map uses a 40 000 ms tile-load wait to ensure satellite tiles render fully before screenshotting; chart screenshots use a 10 ms wait (charts render client-side without external tile requests).

8. Output Files

All outputs are written to the directory defined by ECOSCOPE_WORKFLOWS_RESULTS.

File	Format	Content
overall_rainfall_station_readings.gpkg	GeoPackage	Total rainfall per station, with point geometry
monthly_rainfall_station_readings.gpkg	GeoPackage	Total rainfall per station per month, with point geometry
ndvi_.csv	CSV	NDVI time-series with historic min/max/mean per ranch
rainfall_readings_over_time.html	HTML	Interactive overall rainfall bar chart
monthly_rainfall_readings.html	HTML	Interactive monthly stacked bar chart
rainfall_map.html	HTML	Interactive proportional-symbol rainfall map
_ndvi.html	HTML	Interactive NDVI historic time-series per ranch
rainfall_readings_over_time.png	PNG	Screenshot of overall bar chart (2× resolution)
monthly_rainfall_readings.png	PNG	Screenshot of monthly stacked bar chart (2× resolution)
rainfall_map.png	PNG	Screenshot of rainfall map (2× resolution)
_ndvi.png	PNG	Screenshot of NDVI chart per ranch (2× resolution)
big_life_weather_report.docx	Word	Final populated weather report

9. Workflow Execution Logic

9.1 Skip Conditions

Every task carries two default skip conditions via task-instance-defaults:

- **any_is_empty_df** — if any input DataFrame is empty the task and all its dependants are skipped rather than raising an error.
- **any_dependency_skipped** — if an upstream task was skipped, all downstream tasks that depend on it are also skipped automatically.

get_events_data carries an explicit skipif block with both conditions. Note, however, that raise_on_empty: true means EarthRanger will return an error (not an empty DataFrame) when no events are found — distinguishing a workflow skip from a genuine data-availability error.

persist_ndvi_data has skipif: conditions: [never], ensuring NDVI CSVs are always written even if the NDVI DataFrame might otherwise trigger a skip.

9.2 mapvalues Fan-out (NDVI per Ranch)

The NDVI pipeline fans out per ranch via mapvalues. split_groups partitions the conservancy GeoDataFrame by R_NAME. Each subsequent task in the chain (calculate_ndvi_range, persist_ndvi_data, draw_ndvi, get_ranch_name, persist_ndvi, convrt_ranch_png) receives the argvalues list from its upstream task, processing one ranch per invocation.

zip_groupbykey aligns the per-ranch NDVI chart HTML with the corresponding ranch name string before persist_text, so each output file is named with the correct conservancy suffix.

9.3 Screenshot Configuration

Task	wait_for_timeout	max_concurrent_pages	Reason
convrt_ranch_png	10 ms	3	NDVI charts are self-contained HTML; fast render, parallelised
convrt_br_png	10 ms	1	Bar chart is self-contained HTML
convrt_monthly_png	10 ms	1	Stacked bar chart is self-contained HTML
convert_map_png	40 000 ms	1	Rainfall map loads external tile layers — long wait ensures tiles render before screenshot

10. Software Versions

Package	Version	Role
ecoscope-workflows-core	0.22.17.*	Core task library and workflow engine
ecoscope-workflows-ext-ecoscope	0.22.17.*	Ecoscope spatial analysis tasks (events, map layers, charts)

Package	Version	Role
ecoscope-workflows-ext-custom	0.0.39.*	Custom utility tasks (column transforms, geometry cleaning)
ecoscope-workflows-ext-ste	0.0.17.*	STE-specific tasks (NDVI range, historic timeseries chart)
ecoscope-workflows-ext-mnc	0.0.7.*	MNC domain tasks
ecoscope-workflows-ext-mep	0.12.0.*	MEP tasks (weather report generation, stacked bar chart)
ecoscope-workflows-ext-big-life	0.0.9.*	Big Life Foundation tasks (weather report template context)

Packages are distributed via the prefix.dev conda channel and pinned to patch-compatible versions (.* suffix). The runtime environment is managed by **pixi**.